

Simultaneous Convolution Divisors of Fabius-Type Laws

A prime-adic Hall criterion, finite certificates,
linear rigidity, and a sharp regularity transition

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Abstract

Let $X_p = \sum_{k \geq 0} p^{-k} U_k$, where p is prime and the U_k are independent uniform random variables on $[-1, 1]$. We classify every convolution factor of $\mathcal{L}(X_p)$ that is itself a finite or countable convolution of centered uniform laws. The factor $\mathcal{L}(\sum_j a_j U_j)$ exists if and only if $a_j = 1/n_j$ for positive integers n_j and

$$\#\{j : v_p(n_j) \leq K\} \leq K + 1 \quad (K \geq 0).$$

For this class, cancellation of all poles in the quotient of entire characteristic functions is already sufficient for positivity. A nested matching argument gives an explicit independent residual and finite obstruction certificates. For finitely many geometric streams, the infinitely many inequalities reduce to one finite window and a rational density condition.

In several dimensions, every uniform generator of a convolution factor of an independent product law must lie on a coordinate axis. Consequently, its complete linear decomposability semigroup consists of partial monomial matrices whose nonzero entries are signed reciprocal integers. The scalar matching theorem extends to prime-power bases, but not to arbitrary composite bases: an explicit base-six quotient is entire and is the transform of a signed, not positive, measure.

For pure prime-power scales, the residual has a canonical digit representation whose multiplicities are exactly the unused matching capacities. Positive capacity density gives a compactly supported smooth density, a sharp log-squared Fourier limsup, and differential non-finiteness. At zero capacity density, an eventually periodic digit profile gives a complete classification into finite atomic laws, splines, singular continuous laws, or densities of an explicitly determined exact finite differentiability order. All mathematical assertions designated as theorems are proved below; a standard-library Python companion checks finite certificates and examples.

Status and scope. This is a proof-bearing research manuscript, not a report of a Lean verification or peer review. It develops extensions of the repository's already existing reciprocal-integer *single-copy* divisor theorem, rather than claiming that result as new. The source review was targeted, not an exhaustive audit of every repository artifact or the entire literature. Global priority for the extensions proved here is not asserted.

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1 The question and its relationship to the repository

The Fabius and Rvachev functions provide a particularly concrete meeting point of probability, entire functions, arithmetic, and formal mathematics. Their Fourier transforms are infinite products of sinc functions. A zero of such a product is not just a point where a transform vanishes: its multiplicity records how many geometric scales coincide at that frequency. The question addressed here is whether that arithmetic record completely determines which independent uniform components can be removed from the distribution.

The existing ProveIt report *Reciprocal-Integer Convolution Divisors of the Rvachev Law* already constructs and classifies decompositions of the form $X \stackrel{d}{=} cX' + Y$, where X' is an independent copy of the same Rvachev law [4]. Its reciprocal-integer classification, quotient cocycle, and odd/even regularity results are prior work relative to this manuscript. The repository also contains exact arithmetic descriptions of the zero divisor in `GeneralizedZeroDivisor.lean` [6]. Those results suggest a strictly larger problem:

Which *collections* of scaled uniform laws, or of geometric uniform convolutions with different bases, can occur simultaneously as independent components of a fixed Fabius-type law?

The distinction is substantial. Two factors can each be individually admissible but impossible to remove together. Conversely, several different geometric streams can interleave exactly, or leave a singular Cantor residual rather than a smooth function. Scalar variance inequalities do not detect these phenomena.

The main contribution is an exact answer for prime and prime-power target bases. The answer has four interchangeable forms: convolution divisibility, analytic zero cancellation, a family of arithmetic inequalities, and a constructive matching of source factors to target scales. The matching is a special nested instance of the representative-selection principle associated with Hall’s theorem [2]; its proof here requires only sorting integer deadlines.

Editorial note (ProveIt, 2026-09-29). The repository already contained, at both pinned snapshots, the article `spectra-and-arithmetic/Arithmetic_Convolution_Factors_Fabius_Type_Laws/` (filed 2026-09-28), which this manuscript does not cite. Its theorems “Arithmetic convolution classification” ($\mu_A = D_c \mu_B * \nu$ iff $c = 1/m$ and $A_k \mid mB_k$) and “Cross-base classification” contain the one-stream case of theorem 4.1; its periodic-ratio decision theorem and its integer regularity classification are one-stream cases of theorem 4.2 and theorem 9.3. That article is more general in another direction: it allows composite divisibility-ladder targets, whereas the results here need prime or prime-power geometric targets. Conversely, theorem 3.2, theorem 6.1 and theorem 6.2 answer that article’s question “Beyond divisibility ladders” for prime-power geometric targets (the base-six quotient is the counterexample it asks for), and theorems 5.1 and 5.3 treat the product-target case of its question “Multivariate arithmetic factors”. Its uniforms live on $[-1/2, 1/2]$ and its sinc is normalized, so its law $\mu^{[p]}$ is the law of $X_p/2$ here. Editorial notes at both questions there record these answers. The identity $X_2 \stackrel{d}{=} X_4 + \frac{1}{2}X'_4$ of the example “Density is necessary but not sufficient” is machine-checked, in the $[0, 1]$ -digit normalization, as `Fabius.ProbabilityRepresentation.geometricUniformDistribution_one_half_conv_one_quarter (Analysis/FabiusFunction/Lean/FabiusFunction/GeometricUniformMultisection.lean)`.

1.1 What is established here, and what is reused

Part of the argument	Role and status
Infinite sinc product and its zeros	Established background; proved again in the normalization used here.
Single scaled independent copy	Existing repository result; recovered as a corollary, not claimed as a new contribution.
Simultaneous uniform factors	Complete classification and explicit residual construction in section 3.
Geometric streams	Necessary integer-base restriction and a complete finite decision criterion in section 4.
Multidimensional factors	Coordinate-direction rigidity and the complete matrix classification in section 5.
Analytic versus positive factorization	Equivalence for prime-power targets and a separating base-six example in section 6.
Residual regularity	Exact subcritical Fourier limsup and critical periodic classification in sections 8 and 9.

The terminology of a linear decomposability or Urbanik semigroup is conventional; Jurek describes its relation to generalized Mehler semigroups [3]. The matrix classification below is proved directly and does not invoke an external semigroup-classification theorem.

1.2 A normalization warning

We use uniforms on $[-1, 1]$ and the characteristic-function convention $\widehat{\mu}(t) = \int e^{itx} d\mu(x)$. Thus X_2 is supported on $[-2, 2]$. The classical Rvachev up-density on $[-1, 1]$ is the density of $X_2/2$. Under the repository's convention $\int e^{2\pi izx} d\mu(x)$, the transform of $X_2/2$ is $\prod_{k \geq 0} \text{sinc}(\pi z/2^k)$, as in the classical account [1]. Every scalar decomposition theorem is unchanged by this common normalization of source and target; explicit supports and digit locations are not.

2 Probability and analytic preliminaries

For $a > 0$, let U_a denote the uniform law on $[-a, a]$, and put

$$\text{sinc } z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

If $b > 1$ is real and the U_k are independent with law U_1 , define

$$X_b = \sum_{k=0}^{\infty} b^{-k} U_k, \quad \mu_b = \mathcal{L}(X_b), \quad \Phi_b(z) = \prod_{k=0}^{\infty} \text{sinc}(z/b^k). \quad (1)$$

The series converges absolutely for every outcome. Its support is the interval $[-b/(b-1), b/(b-1)]$: finite sums have interval support with radii tending to $b/(b-1)$, and the remaining summands have radii tending to zero. Independence and bounded convergence give $\widehat{\mu_b}(t) = \Phi_b(t)$. In particular,

$$\mathbb{E}X_b = 0, \quad \text{Var}(X_b) = \frac{1}{3} \sum_{k \geq 0} b^{-2k} = \frac{b^2}{3(b^2 - 1)}. \quad (2)$$

For a finite or countable indexed family $a = (a_j)$ of positive numbers with $\sum_j a_j < \infty$, let

$$\nu_a = \mathcal{L} \left(\sum_j a_j U_j \right), \quad \Psi_a(z) = \prod_j \text{sinc}(a_j z). \quad (3)$$

The empty family gives the point mass δ_0 and the product 1. All copies of uniforms or other random variables occurring in an asserted independent sum are understood to be mutually independent.

Definition 2.1 (Convolution divisibility). *A probability law ν divides a probability law μ , written $\nu \mid_* \mu$, if there is a probability law ρ such that $\mu = \nu * \rho$. The residual ρ is not assumed to belong to any prescribed family. For a scalar c , $D_c \nu$ denotes the pushforward of ν by $x \mapsto cx$.*

Lemma 2.2 (Entire products and exact zeros). *The products in (1) and (3) converge locally uniformly on $\mathbb{C}$ and are entire. They have precisely the zeros contributed by the individual factors, with multiplicities added. For prime p ,*

$$Z(\Phi_p) = \{\pi m : m \in \mathbb{Z} \setminus \{0\}\}, \quad \text{ord}_{\pi m} \Phi_p = 1 + v_p(|m|). \quad (4)$$

Here $v_p(n)$ is the exponent of p in a positive integer n .

Proof. On a fixed compact set, $\text{sinc}(a_j z) = 1 + O(a_j^2)$ uniformly once a_j is small. Since $\sum a_j^2 < \infty$, the product converges locally uniformly. Away from zeros of the finitely many large factors, a sufficiently far tail admits a uniformly absolutely convergent logarithm and hence is nonzero. At any fixed nonzero z , only finitely many factors can vanish, because a vanishing factor requires $a_j |z| \geq \pi$. The zeros of every individual sinc factor are simple. These observations prove the first assertion and the multiplicity rule.

For $a_k = p^{-k}$, a zero is $\pi p^k \ell$ with ℓ a nonzero integer. Their union is exactly the nonzero integer multiples of π . At πm , the vanishing factors are those with $p^k \mid m$, namely $k = 0, \dots, v_p(|m|)$. $\square$

Lemma 2.3 (Compact residuals and uniqueness). *If compactly supported probability laws satisfy $\mu = \nu * \rho$, then ρ is compactly supported. If $\hat{\nu}$ extends to a nonzero entire function, the residual is unique. In particular, $\nu_a \mid_* \mu_b$ implies that Φ_b / Ψ_a has an entire extension.*

Proof. For $x \in \text{supp } \nu$ and $y \in \text{supp } \rho$, every sufficiently small neighborhood of $x + y$ receives positive convolution mass from neighborhoods of x and y . Thus $x + y \in \text{supp } \mu$. Fixing one $x \in \text{supp } \nu$ confines $\text{supp } \rho$ to a translate of the compact set $\text{supp } \mu$. The characteristic function of a compactly supported law is entire, by differentiation under a bounded-support integral. The convolution identity extends from the real line to $\mathbb{C}$ by the identity theorem, proving the quotient assertion.

The real zeros of a nonzero entire function are isolated. Hence two residual characteristic functions agree on a dense subset of $\mathbb{R}$ by cancellation, and then everywhere by continuity. Uniqueness of characteristic functions gives uniqueness of the probability laws. $\square$

3 A complete prime-adic Hall criterion

Each target uniform $p^{-k} U_k$ will be called supply slot k . A source uniform U/n can be extracted from that slot whenever $p^k \mid n$. The crucial feature is that the eligible slots are a nested initial interval, not an arbitrary set.

Lemma 3.1 (Splitting one uniform). *If $n, r \in \mathbb{N}$ and $a = r/n$, then*

$$U_a = U_{1/n} * \mathcal{L}(V_{n,r}), \quad \mathbb{P}\left(V_{n,r} = \frac{2\ell - r + 1}{n}\right) = \frac{1}{r} \quad (0 \leq \ell < r). \quad (5)$$

For $r = 1$ the digit law is δ_0 .

Proof. The r translates of $[-1/n, 1/n]$ centered at the displayed points tile $[-r/n, r/n]$, with overlaps only at endpoints. Each mixture component has weight $1/r$ and density $n/2$, so the combined density is $1/(2a)$ on that interval and zero outside it. Notice also that $|V_{n,r}| \leq a - 1/n$. $\square$

Theorem 3.2 (Complete simultaneous-factor classification). *Let p be prime and let ν_a be as in (3). The following conditions are equivalent.*

- (i) $\nu_a \mid_* \mu_p$.
- (ii) Φ_p/Ψ_a extends to an entire function on $\mathbb{C}$.
- (iii) There are positive integers n_j such that $a_j = 1/n_j$ and

$$D(K) := \#\{j : v_p(n_j) \leq K\} \leq K + 1 \quad \text{for every } K \in \mathbb{N}_0. \quad (6)$$

- (iv) There are positive integers $n_j = 1/a_j$ and an injection $j \mapsto k_j$ into $\mathbb{N}_0$ such that $p^{k_j} \mid n_j$ for every j .

When these conditions hold, the residual is unique and has an explicit representation as an absolutely convergent sum of independent finite digit variables and the unmatched target uniforms.

Proof. (i) $\Rightarrow$ (ii). This is theorem 2.3.

(ii) $\Rightarrow$ (iii). The first positive zero π/a_j of any source factor must be a zero of Φ_p . By (4), $1/a_j$ is a positive integer, denoted n_j .

Suppose (6) fails at K . Choose $K + 2$ source indices whose $v_p(n_j)$ are at most K and put m equal to the least common multiple of these denominators. Then $v_p(m) \leq K$. Every chosen factor vanishes at πm , so Ψ_a has order at least $K + 2$ there, whereas Φ_p has order at most $K + 1$. The quotient has a pole, a contradiction. This argument also handles an infinite set of demands at a fixed deadline: a finite subfamily suffices.

(iii) $\Rightarrow$ (iv). Write $d_j = v_p(n_j)$. Condition (6) makes each bounded-deadline set finite. Therefore the source indices can be enumerated in nondecreasing deadline order, with arbitrary fixed ordering of ties. Let $d_{(k)}$ be the deadline of the zero-indexed k th source in this enumeration. If $d_{(k)} < k$, the first $k + 1$ sources would all have deadline at most $d_{(k)}$, contradicting $D(d_{(k)}) \leq d_{(k)} + 1$. Thus $d_{(k)} \geq k$. Assign the k th source to slot k . This is the required injection. For a finite family only the corresponding finite initial set of slots is used.

(iv) $\Rightarrow$ (iii). Every source of deadline at most K must be assigned to one of slots $0, \dots, K$. Injectivity gives (6).

(iv) $\Rightarrow$ (i). For the source assigned to slot k_j , put $r_j = n_j/p^{k_j} \in \mathbb{N}$. By theorem 3.1,

$$p^{-k_j} U \stackrel{d}{=} n_j^{-1} U' + V_{n_j, r_j}.$$

Use this splitting independently at every matched slot. Keep the entire uniform $p^{-k} U_k$ at every unmatched slot. All the residual summands have absolute value at most p^{-k} at their respective slots, so their sum converges absolutely. The extracted source variables also converge absolutely. Their independence and the slot-by-slot identity give $\mu_p = \nu_a * \rho$. Uniqueness follows from theorem 2.3. $\square$

Remark 3.3 (Why positivity is a conclusion). An entire quotient of two characteristic functions need not be a characteristic function. In theorem 3.2, positivity follows from the explicit independent construction, not from the analytic cancellation by itself. The arithmetic structure allows the cancellation condition to force a valid construction. Section 6 gives an entire quotient for which positivity fails when that structure is absent.

Remark 3.4 (Summability is also forced by the certificate). If a countable family of positive integers satisfies (6), its sorted enumeration has $n_{(k)} \geq p^k$. Consequently $\sum_j 1/n_j \leq p/(p-1)$. Thus the arithmetic condition itself supplies the absolute-convergence assumption needed to define the source probability law.

3.1 Finite witnesses and exact residual bounds

Corollary 3.5 (A finite certificate for every failure). *For reciprocal-integer widths, failure of convolution divisibility has a finite witness: a finite subfamily whose least common multiple m satisfies*

$$\#\{j \text{ in the subfamily} : n_j \mid m\} > 1 + v_p(m).$$

For a finite input family, sorting deadlines and testing $d_{(k)} \geq k$ either constructs the residual slot matching or produces such a witness.

Proof. If $d_{(k)} < k$, take the first $k+1$ sorted sources. The valuation of their least common multiple is $d_{(k)}$, so the source order exceeds the target order. Conversely, such an excess prevents an entire quotient by theorem 2.2. The algorithm terminates after the finite sort and scan. $\square$

Proposition 3.6 (Support endpoints, variance, and a truncation certificate). *For an admissible source, put*

$$R = \frac{p}{p-1} - \sum_j \frac{1}{n_j}.$$

The residual is symmetric, is supported in $[-R, R]$, and has both $-R$ and R in its support. Its variance is

$$\text{Var}(\rho) = \frac{1}{3} \left(\frac{p^2}{p^2-1} - \sum_j \frac{1}{n_j^2} \right). \quad (7)$$

If $\rho^{[K]}$ retains only residual summands assigned to target slots $0, \dots, K-1$, then

$$W_\infty(\rho, \rho^{[K]}) \leq \frac{p^{1-K}}{p-1}. \quad (8)$$

Here W_∞ is the infimum of the essential maximum displacement over all couplings. No assertion that the residual support is an interval is intended.

Proof. Each split digit law is symmetric and has extreme support points $\pm(p^{-k_j} - n_j^{-1})$. Unmatched slots have extreme points $\pm p^{-k}$. Summing these radii gives R . Every finite set of summands can simultaneously lie arbitrarily near their upper endpoints with positive probability, and the remaining total radius tends to zero. Thus R , and similarly $-R$, belongs to the support. Variances subtract in the independent decomposition and yield (7). Finally, couple the full residual and its truncation using the same retained summands. The discarded absolute displacement is at most $\sum_{k \geq K} p^{-k} = p^{1-K}/(p-1)$. $\square$

4 Geometric streams: an infinite problem with a finite certificate

We next allow finitely many entire Fabius-type laws as source components. A geometric stream means the sequence of uniforms making up one scaled copy of X_b . Besides these streams, finitely many individual uniform factors may be present.

Theorem 4.1 (Classification of geometric source families). *Fix prime p . Consider the law of*

$$Y = \sum_{\ell=1}^J u_\ell U_\ell + \sum_{i=1}^r c_i X_{b_i}^{(i)}, \quad u_\ell, c_i > 0, \quad b_i > 1, \quad (9)$$

with all components independent. It divides μ_p if and only if all of the following hold:

- (a) $u_\ell = 1/N_\ell$ and $c_i = 1/n_i$ for positive integers N_ℓ, n_i ;
- (b) each b_i is an integer divisible by p ;
- (c) writing $d_\ell = v_p(N_\ell)$, $a_i = v_p(n_i)$, and $s_i = v_p(b_i) \geq 1$, one has

$$D(K) = \sum_{\ell=1}^J \mathbf{1}_{d_\ell \leq K} + \sum_{i=1}^r \max \left\{ 0, 1 + \left\lfloor \frac{K - a_i}{s_i} \right\rfloor \right\} \leq K + 1 \quad (K \geq 0). \quad (10)$$

Negative nonzero coefficients are handled by replacing them with their absolute values, since all source laws are symmetric. Zero coefficients may be discarded.

Proof. Expand every X_{b_i} into its uniform summands and apply theorem 3.2. The first term of each stream forces $c_i = 1/n_i$, and all stream denominators $n_i b_i^j$ must be integers. The case $j = 1$ makes b_i rational. Write $b_i = u/v$ in lowest terms. Since $n_i u^j / v^j$ is an integer for every j , v^j divides n_i for every j , and therefore $v = 1$. Thus b_i is an integer. If p did not divide b_i , the infinitely many denominators of this stream would all have the same p -adic valuation, violating (6).

For an integer base divisible by p , the deadlines of stream i are exactly $a_i + s_i j$ for $j \geq 0$. Counting how many are at most K gives (10). The converse follows by the same expansion and the sufficiency part of theorem 3.2. $\square$

Define

$$A = \max(\{0\} \cup \{d_\ell\} \cup \{a_i\}), \quad L = \text{lcm}(s_1, \dots, s_r), \quad \eta = \sum_{i=1}^r \frac{1}{s_i}, \quad \delta = 1 - \eta. \quad (11)$$

For no streams, the conventions are $L = 1$ and $\eta = 0$. Put $F(K) = K + 1 - D(K)$. The parameter δ is the asymptotic surplus of supply slots over source demands per valuation level.

Theorem 4.2 (Finite-window criterion). *All inequalities in (10) hold if and only if*

$$\delta \geq 0 \quad \text{and} \quad F(K) \geq 0 \quad \text{for } 0 \leq K < A + L. \quad (12)$$

More precisely, for every $K \geq A$,

$$F(K + L) = F(K) + L\delta, \quad L\delta = L - \sum_i L/s_i \in \mathbb{Z}. \quad (13)$$

Proof. For $K \geq A$, every finite demand has already arrived and every stream has started. Increasing K by L adds exactly L/s_i demands from stream i and adds L supply slots. This proves (13). If $L\delta < 0$, iterating along any residue class eventually makes F negative. If $L\delta \geq 0$, every $K \geq A$ is obtained from a unique element of $[A, A + L - 1]$ by adding a nonnegative multiple of L , and its slack cannot decrease. The earlier indices are precisely the remaining finite checks in (12). $\square$

Algorithmic interpretation. Compute the valuations, A , L , and the integer drift $L\delta$. Scan the $A + L$ displayed inequalities. A negative entry is a finite rejection certificate. If all pass but the drift is negative, let $q = \lfloor F(A)/(-L\delta) \rfloor + 1$; then $K = A + qL$ is an explicit violating index. If all pass and the drift is nonnegative, (13) certifies all remaining indices. The method is exact, but its least-common-multiple window need not be polynomial in the binary length of the input. An implementation must distinguish a resource limit from a negative mathematical certificate.

Example 4.3 (Density is necessary but not sufficient). *For $p = 2$, two unscaled copies of X_4 have deadline streams $0, 2, 4, \dots$. Their combined density is $1/2 + 1/2 = 1$, but they already demand two slots at $K = 0$, where only one is available. In contrast, the two streams $0, 2, 4, \dots$ and $1, 3, 5, \dots$ use every slot once and give*

$$X_2 \stackrel{d}{=} X_4 + \frac{1}{2}X'_4.$$

The simultaneous criterion distinguishes these cases before any analytic approximation or numerical positivity test is needed.

Corollary 4.4 (The existing single-copy theorem, recovered). *For $c \neq 0$, $D_c\mu_p \mid_* \mu_p$ if and only if $|c| = 1/n$ for a positive integer n . Including $c = 0$ gives the trivial factor δ_0 .*

Proof. A single stream with base p has step $s = 1$ and start $a = v_p(n) \geq 0$; its demand is $\max(0, K - a + 1) \leq K + 1$. Necessity is already part of theorem 4.1. For $p = 2$, this recovers the classification established in the repository's divisor report [4]. $\square$

5 Multidimensional rigidity and the matrix semigroup

The scalar classification has a strong geometric consequence. Let $d \geq 1$ and consider the target product law $\mu_p^{\otimes d}$ on $\mathbb{R}^d$. Its characteristic function is

$$\Phi_{p,d}(t) = \prod_{q=1}^d \Phi_p(t_q).$$

Its real zero set is a countable union of coordinate hyperplanes. A source uniform along an oblique line creates an oblique zero hyperplane, and there is no positive residual that can remove it.

Theorem 5.1 (Rigidity of uniform generators). *Let $v_j \in \mathbb{R}^d \setminus \{0\}$ be a finite or countable family with $\sum_j \|v_j\| < \infty$, and let $Y = \sum_j v_j U_j$. Then $\mathcal{L}(Y) \mid_* \mu_p^{\otimes d}$ if and only if every generator has the form*

$$v_j = \varepsilon_j \frac{e_{q(j)}}{n_j}, \quad \varepsilon_j \in \{-1, 1\}, \quad n_j \in \mathbb{N}, \quad (14)$$

and each coordinate separately satisfies

$$\#\{j : q(j) = q, v_p(n_j) \leq K\} \leq K + 1 \quad (q = 1, \dots, d; K \geq 0). \quad (15)$$

The residual is unique and is a product of its coordinate residual laws.

Proof. Suppose first that the factorization exists. Fix a generator $v = v_j$. The source characteristic function vanishes everywhere on $H = \{t \in \mathbb{R}^d : v \cdot t = \pi\}$. Thus the target must vanish there, and

$$H \subseteq \bigcup_{q=1}^d \bigcup_{m \in \mathbb{Z} \setminus \{0\}} \{t : t_q = \pi m\}.$$

If v has at least two nonzero coordinates, no coordinate hyperplane in the union contains H . Its intersection with H is consequently either empty or a proper affine subspace of H , hence closed and nowhere dense in H . The affine space H is complete and cannot be a countable union of such subsets by the Baire category theorem. This contradiction proves that every v_j lies on a coordinate axis. When $d = 1$, that conclusion is automatic.

After grouping generators by coordinate and absorbing their signs by symmetry, the coordinate projections of the purported factorization are scalar factorizations of μ_p . Apply theorem 3.2 to each projection. It gives (14) and (15).

Conversely, the scalar construction supplies a residual in every coordinate. Taking their independent product gives the required vector decomposition. Finally, the source characteristic function is nonzero on a dense subset of $\mathbb{R}^d$: it is a product of one-variable entire functions with isolated real zeros. Thus cancellation and continuity make the residual characteristic function unique. It must be the product already constructed. $\square$

Definition 5.2 (Linear decomposability semigroup). *For a probability law μ on $\mathbb{R}^d$, define*

$$\mathcal{D}(\mu) = \{C \in \mathbb{R}^{d \times d} : \mu = (C_*\mu) * \rho \text{ for some probability law } \rho\}.$$

This is a semigroup under matrix multiplication: combining factorizations for C and D gives a factorization for CD . A partial monomial matrix has at most one nonzero entry in each row and at most one in each column.

Theorem 5.3 (Complete matrix classification). *For prime p and every $d \geq 1$,*

$$\mathcal{D}(\mu_p^{\otimes d}) = \left\{ C : C \text{ is partial monomial and every nonzero entry is } \frac{\varepsilon}{n}, \varepsilon \in \{-1, 1\}, n \in \mathbb{N} \right\}. \quad (16)$$

The invertible matrices in this set are signed permutation matrices with reciprocal-integer coordinate scalings. Its semigroup units are exactly the signed permutation matrices. Every continuous path taking values in $\mathcal{D}(\mu_p^{\otimes d})$ is constant.

Proof. Let X' have independent coordinates with law μ_p . Writing each coordinate as its uniform series expresses CX' using the generators $p^{-k}c_j$, where c_j is column j of C . By theorem 5.1, every nonzero column is coordinate-aligned and its coefficient is a signed reciprocal integer.

If two nonzero columns were assigned to the same row, that row would contain two deadline streams, each of step 1. Their combined demand is $2K + O(1)$, which eventually exceeds $K + 1$. Hence each row contains at most one nonzero column, proving the necessary partial monomial structure. Conversely, every nonzero row of such a matrix invokes exactly one admissible scalar single-copy factor from theorem 4.4; zero rows leave a full target coordinate as residual. The independent rowwise constructions prove sufficiency.

Invertibility forces exactly one nonzero entry in each row and column. The inverse can also satisfy (16) only when every scaling integer is 1, proving the assertion about units. Finally, each matrix entry of a continuous path lies in the countable subset $\{0\} \cup \{\pm 1/n : n \in \mathbb{N}\}$ of $\mathbb{R}$. A connected subset of $\mathbb{R}$ with more than one point contains an interval and is uncountable. Each entry, and hence the path, must therefore be constant. $\square$

Corollary 5.4 (No continuous nontrivial linear scaling family). *There is no continuous nonconstant family $C_t \in \mathcal{D}(\mu_p^{\otimes d})$ with $C_0 = I$. In particular, no continuous linear contraction semigroup tending to zero can provide decompositions of this fixed target law at every time.*

Proof. Apply the final assertion of theorem 5.3. $\square$

This is a rigidity statement about *exact independent decompositions*; it is not a prohibition on approximate decompositions, nonlinear dynamics, or unrelated stochastic processes. It

also explains why the scalar reciprocal integers should not be mistaken for a continuous self-decomposability property. With different source bases, several columns may feed the same output coordinate, but the corresponding row must then satisfy the stream criterion of theorem 4.1.

6 Prime-power targets and a composite-base obstruction

Theorem 6.1 (Prime-power extension). *Let $B = p^h$ with p prime and $h \geq 1$. For a summable uniform source ν_a , convolution divisibility $\nu_a \mid_* \mu_B$ and entire extensibility of Φ_B/Ψ_a are equivalent. They hold exactly when $a_j = 1/n_j$ with $n_j \in \mathbb{N}$ and*

$$\# \left\{ j : \left\lfloor \frac{v_p(n_j)}{h} \right\rfloor \leq K \right\} \leq K + 1 \quad (K \geq 0). \quad (17)$$

The residual construction assigns source j to a distinct slot k_j with $B^{k_j} \mid n_j$ and splits that uniform as in theorem 3.1.

Proof. The target zeros are still precisely πm , $m \neq 0$, but their orders are

$$1 + \left\lfloor \frac{v_p(|m|)}{h} \right\rfloor.$$

The allowed slots of denominator n are the initial interval through $\lfloor v_p(n)/h \rfloor$. The least-common-multiple obstruction works because

$$\left\lfloor \frac{v_p(\text{lcm}(n_1, \dots, n_s))}{h} \right\rfloor = \max_i \left\lfloor \frac{v_p(n_i)}{h} \right\rfloor.$$

The necessity, sorted matching, and independent splitting proofs of theorem 3.2 therefore apply word for word with these deadlines and supply scales B^{-k} . $\square$

For a general integer B , the eligible supply slots of a single denominator are still nested. What fails is the least-common-multiple reduction: different prime factors can be supplied by different denominators and can create extra powers of B in their least common multiple. The resulting analytic zero condition can miss a genuine positivity obstruction.

Theorem 6.2 (An entire nonpositive quotient at base six). *The function*

$$Q(t) = \frac{\Phi_6(t)}{\text{sinc}(t/2) \text{sinc}(t/3)} \quad (18)$$

*extends to an entire function, is normalized by $Q(0) = 1$, and is not the characteristic function of a probability measure. In particular, $\mathbf{U}_{1/2} * \mathbf{U}_{1/3}$ does not divide μ_6 .*

Proof. Peeling off the first two factors of Φ_6 and using elementary trigonometric identities gives

$$\begin{aligned} Q(t) &= \frac{\text{sinc } t \text{ sinc}(t/6)}{\text{sinc}(t/2) \text{sinc}(t/3)} \Phi_6(t/36) \\ &= \frac{\cos(t/2)}{\cos(t/6)} \Phi_6(t/36) \\ &= (2 \cos(t/3) - 1) \Phi_6(t/36). \end{aligned} \quad (19)$$

The identities first hold away from their displayed denominators' zeros and then extend analytically. The final expression is entire.

Let $\tau = D_{1/36}\mu_6$. Its support is $[-1/30, 1/30]$. Equation (19) is the characteristic transform of the finite signed measure

$$(\delta_{-1/3} + \delta_{1/3} - \delta_0) * \tau.$$

The three translates have disjoint containing intervals. On the central interval $[-1/30, 1/30]$, this signed measure has mass exactly -1 . It is therefore not positive. Uniqueness of Fourier transforms for finite signed measures prevents a different positive measure from having the same transform. Finally, an actual convolution residual in (18) would have characteristic function Q , contradicting what was just proved. $\square$

The example has an analytic quotient with every pole canceled, so it cannot be rejected by searching for an unremoved Fourier zero. It shows precisely why a positivity construction is indispensable and why the prime-power theorem is not a routine assertion about arbitrary entire quotients.

7 The residual digit profile

From now until explicitly stated otherwise, the target base is prime p and the source denominators are pure powers $n_j = p^{d_j}$ with $d_j \in \mathbb{N}_0$. Assume the Hall inequalities hold. Define the *digit slack*

$$g_h = h - \#\{j : d_j < h\} = F(h-1), \quad h \geq 1. \quad (20)$$

In particular, $0 \leq g_h \leq h$. This is the number of target digit layers remaining at level h after all source digit layers are removed.

Let D_p be uniform on the centered digit set $\{2\ell - p + 1 : 0 \leq \ell < p\}$ and write

$$\chi_p(u) = \mathbb{E}e^{iuD_p} = \frac{1}{p} \sum_{\ell=0}^{p-1} e^{i(2\ell-p+1)u} = \frac{\sin(pu)}{p \sin u}. \quad (21)$$

The quotient has its removable values at multiples of π ; the finite sum is its globally valid definition.

Lemma 7.1 (Uniform digit expansion). *For every complex z ,*

$$\text{sinc } z = \prod_{h=1}^{\infty} \chi_p(z/p^h). \quad (22)$$

Equivalently, $\sum_{h \geq 1} p^{-h} D_{p,h}$ has law U_1 for independent copies $D_{p,h}$ of D_p .

Proof. The finite product telescopes to $\sin z / [p^N \sin(z/p^N)]$, with removable values understood. Its locally uniform limit is $\text{sinc } z$. The digit series is absolutely convergent, and its characteristic function is the product on the right. Uniqueness of characteristic functions proves the probability statement. $\square$

Theorem 7.2 (Canonical residual digit law). *For the pure-power source above, the unique residual has the representation*

$$Z = \sum_{h=1}^{\infty} p^{-h} \sum_{\ell=1}^{g_h} D_{h,\ell}, \quad \widehat{\rho}(t) = \prod_{h=1}^{\infty} \chi_p(t/p^h)^{g_h}, \quad (23)$$

where the $D_{h,\ell}$ are independent with law D_p . The series is absolutely convergent. Moreover,

$$\text{ord}_{\pi p^\kappa} \widehat{\rho} = g_{K+1} = F(K) \quad (K \geq 0). \quad (24)$$

Proof. Expand every uniform using theorem 7.1. The target product has exponent h at digit level h : the pairs consisting of supply scale $k \geq 0$ and a positive relative digit index with sum h number exactly h . A source uniform $p^{-d_j}U$ contributes one digit at every level $h > d_j$. The difference of the two multiplicities is exactly (20).

Since $\sum_h hp^{-h} < \infty$ and $|D_p| \leq p - 1$, all these series and regroupings are absolutely convergent. The source digits plus the digits in (23) reproduce the target digits independently. This proves the factorization and hence, by uniqueness, the claimed residual. The zero-order formula also follows by subtracting the source order $\#\{j : d_j \leq K\}$ from the target order $K + 1$ at πp^K . $\square$

The construction in theorem 3.2 is a matching of *whole uniforms*; theorem 7.2 instead matches their *individual digit layers*. They produce the same law but answer different questions. The first is well suited to an exact existence certificate for arbitrary integer denominators. The second exposes regularity, arithmetic scaling, and entire-function structure.

For a finite-plus-geometric pure-power source, the definitions in (11) give

$$g_h = \delta h + O(1). \quad (25)$$

If $\delta = 0$, (13) says that g_h is periodic with period L for every $h \geq A + 1$. These two regimes lead to very different residuals.

8 Positive surplus: smoothness, sharp Fourier decay, and non-D-finiteness

Theorem 8.1 (A sharp log-squared Fourier envelope). *Let a pure-power source satisfy the Hall condition and suppose its digit slack has $g_h = \delta h + O(1)$ with $\delta > 0$. Then its residual has a compactly supported C^∞ density and*

$$\log |\hat{\rho}(t)| \leq -\frac{\delta}{2 \log p} (\log |t|)^2 + O(\log |t|) \quad (|t| \rightarrow \infty), \quad (26)$$

where $\log 0 = -\infty$. The coefficient is sharp:

$$\limsup_{|t| \rightarrow \infty} \frac{\log |\hat{\rho}(t)|}{(\log |t|)^2} = -\frac{\delta}{2 \log p}. \quad (27)$$

The limsup is attained as a limit along $t_N = \pi p^N / (p + 1)$.

Proof. Choose a nonnegative constant C so that $g_h \geq \delta h - C$ for all h , and define the nondecreasing integer minorant

$$e_h = \max(0, \lfloor \delta h - C \rfloor) \leq g_h.$$

For $j \geq 1$, let H_j be the first index with $e_h \geq j$. Then $H_j = j/\delta + O(1)$, and another application of the uniform digit expansion gives

$$\prod_{h \geq 1} \chi_p(t/p^h)^{e_h} = \prod_{j \geq 1} \text{sinc}(t/p^{H_j-1}). \quad (28)$$

The remaining digit multiplicities $g_h - e_h$ define a probability law, so their characteristic function has modulus at most 1 on the real axis.

Put $T = \log_p |t|$ and retain from (28) the $J = \delta T + O(1)$ factors with $H_j - 1 \leq T$. The elementary bound $|\operatorname{sinc} u| \leq \min(1, 1/|u|)$ yields

$$\begin{aligned} \log |\widehat{\rho}(t)| &\leq -J \log |t| + \log p \sum_{j=1}^J (H_j - 1) \\ &= -J \log |t| + \frac{\log p}{2\delta} J^2 + O(J) \\ &= -\frac{\delta}{2 \log p} (\log |t|)^2 + O(\log |t|). \end{aligned}$$

This proves the upper bound uniformly in real t of large modulus, including zeros with the stated convention.

For sharpness, put $\alpha = \pi/(p+1)$ and $t_N = \alpha p^N$. Since $p \equiv -1 \pmod{p+1}$, for every integer $j \geq 0$,

$$|\sin(\alpha p^j)| = \sin \alpha, \quad |\chi_p(\alpha p^j)| = 1/p.$$

The first N digit factors in (23) therefore contribute

$$-\log p \sum_{h=1}^N g_h = -\frac{1}{2} \delta \log p N^2 + O(N)$$

to the logarithmic modulus. The remaining arguments are α/p^r , $r \geq 1$, on an interval where χ_p is positive and nonzero. Taylor's theorem and compactness give $0 \leq -\log \chi_p(\alpha/p^r) \leq C_1 p^{-2r}$ for a fixed constant C_1 . Since $g_{N+r} = O(N+r)$, the entire tail contributes $O(N)$ in absolute value. Thus

$$\log |\widehat{\rho}(t_N)| = -\frac{1}{2} \delta \log p N^2 + O(N).$$

Dividing by $(N \log p + \log \alpha)^2$ proves the lower bound in (27); the upper bound was already established.

Finally, (26) implies $\int_{\mathbb{R}} |t|^m |\widehat{\rho}(t)| dt < \infty$ for every nonnegative integer m . Fourier inversion supplies a continuous density, and differentiation under the integral supplies continuous derivatives of every order. The law was already compactly supported, so this density is compactly supported as well. $\square$

Corollary 8.2 (Smoothness for arbitrary integer stream denominators). *Every admissible finite-plus-geometric family in theorem 4.1 with $\delta > 0$ has a compactly supported C^∞ residual density. Its characteristic function satisfies the upper bound (26) with the same δ . Sharpness of that coefficient is asserted here only for the pure-power case of theorem 8.1.*

Proof. Replace each source denominator n by $p^{v_p(n)}$, making that source uniform larger. The new family has exactly the same deadlines, is admissible by theorem 3.2, and has a pure-power residual ρ_0 governed by theorem 8.1. Each enlarged source uniform splits into the original uniform and an independent finite digit law, because $n/p^{v_p(n)}$ is an integer. Summing these splits is legitimate: the enlarged widths are summable by the Hall bound. Thus the enlarged source equals the original source convolved with a probability law σ , and the original residual is $\rho_0 * \sigma$. Its Fourier modulus is bounded by $|\widehat{\rho_0}|$. Fourier inversion gives the stated smoothness and upper bound. $\square$

Definition 8.3 (D-finiteness). *An entire function f is D-finite if it satisfies a nontrivial homogeneous linear differential equation $\sum_{j=0}^r P_j(z) f^{(j)}(z) = 0$ with polynomial coefficients P_j .*

Theorem 8.4 (Positive-surplus residuals are not D-finite). *For every admissible finite-plus-geometric source from theorem 4.1 with $\delta > 0$, the entire residual characteristic function is not D-finite. This includes non-pure integer bases and denominators.*

Proof. At valuation cutoff K , the source denominators with p -adic valuation at most K form a finite set of size $D(K)$. Let q_K be the least common multiple of their prime-to- p parts, with the empty least common multiple equal to 1, and put $M_K = p^K q_K$. Exactly those $D(K)$ source factors vanish at πM_K : their denominators divide M_K , while every other denominator has p -adic valuation greater than K . The target order there is $K + 1$. Consequently

$$\text{ord}_{\pi M_K} \hat{\rho} = K + 1 - D(K) = \delta K + O(1),$$

which tends to infinity. These are infinitely many distinct points since $v_p(M_K) = K$.

Suppose a differential equation of order r with leading coefficient $P_r \neq 0$ existed. A nonzero entire function cannot satisfy a nontrivial order-zero equation, so $r \geq 1$. At any point where P_r does not vanish, local uniqueness for the linear analytic differential equation implies that a solution with its first r initial derivatives zero is identically zero. Thus a nonzero solution has zero multiplicity less than r at every such ordinary point. The leading polynomial has only finitely many exceptional points, contradicting the infinite sequence of unbounded multiplicities. The residual is nonzero because $\hat{\rho}(0) = 1$. $\square$

The zero-multiplicity obstruction for the original Rvachev transform is already present in the repository's representation work [7]. The content here is its extension to the residuals, where the remaining zero order is the explicit matching slack and is controlled by the geometric demand density.

9 Critical periodic slack: a complete regularity classification

Assume the pure-power source is admissible and its slack g_h is eventually periodic. This includes every finite-plus-geometric pure-power family with $\delta = 0$, by theorem 4.2. The result below is more general than that application: it only needs eventual periodicity of the actual slack.

The distinction between an atomless singular law and a discrete law matters. Convolving either with a uniform produces an absolutely continuous law, but the singular continuous case gains one more order of continuous differentiability than the discrete case. We first prove the precise singularity statement needed for this distinction.

Lemma 9.1 (A missing periodic digit layer forces singular continuity). *Let $(r_h)_{h \geq 1}$ be a periodic sequence of nonnegative integers. Suppose $\min_h r_h = 0$ and the sequence is not identically zero. Then*

$$S = \sum_{h \geq 1} p^{-h} \sum_{\ell=1}^{r_h} D_{h,\ell} \quad (29)$$

has a singular continuous law: it is singular with respect to Lebesgue measure and has no atoms.

Proof. Replace each centered digit $D_{h,\ell}$ by $2K_{h,\ell} - (p-1)$, where $K_{h,\ell}$ is uniform on $\{0, \dots, p-1\}$. This changes the law by an invertible affine transformation. It therefore suffices to study

$$W = \sum_{h \geq 1} p^{-h} \sum_{\ell=1}^{r_h} K_{h,\ell}.$$

Let L be a period and $B = p^L$. Grouping consecutive blocks of L levels gives

$$W = \sum_{j=0}^{\infty} B^{-(j+1)} A_j, \quad (30)$$

where the A_j are independent, identically distributed, integer-valued finite digits. They need not lie between 0 and $B - 1$; that restriction is not used. Modulo one, multiplication by B deletes the initial block. Consequently the projected law λ on $\mathbb{R}/\mathbb{Z}$ is invariant under $T_B(x) = Bx \pmod{1}$ and is a factor of the independent block shift.

The independent block shift is ergodic. Indeed, sufficiently separated cylinder events are independent; approximation by cylinder events gives mixing for all measurable events. For a shift-invariant event E , this gives $\mathbb{P}(E) = \mathbb{P}(E)^2$. Every invariant event under the factor map pulls back to such an event, so λ is ergodic as well.

Choose $r_0 \in \{1, \dots, L\}$ with $r_{r_0} = 0$, and set $n = p^{r_0-1}$. The Fourier coefficient $\int e^{2\pi i n x} d\lambda(x)$ is a product of digit coefficients. At levels $h < r_0$ all factors are 1. The potentially vanishing factor at level $h = r_0$ is absent. At levels $h > r_0$, the digit ratio is $e^{2\pi i / p^{h-r_0+1}}$, so its average over p consecutive digits is nonzero. The resulting infinite tail product is nonzero as well: the digit multiplicities are bounded and the differences of these factors from 1 are summable. Thus λ has a nonzero Fourier coefficient at the nonzero integer n , and is not Haar measure.

For completeness, an ergodic T_B -invariant probability on $\mathbb{R}/\mathbb{Z}$ that is not Haar must be singular. The map T_B sends absolutely continuous measures to absolutely continuous measures and singular measures to singular measures: for the latter assertion, a null support remains null under the piecewise linear map. Uniqueness of the Lebesgue decomposition therefore makes the absolutely continuous and singular components of an invariant law separately invariant. Ergodicity forces one component to have full mass. One way to see this last assertion directly is to let $f = d\lambda_{ac}/d\lambda$, so $0 \leq f \leq 1$. Invariance of both measures gives

$$\int f(f \circ T_B) d\lambda = \int f^2 d\lambda, \quad \int (f \circ T_B)^2 d\lambda = \int f^2 d\lambda.$$

Hence $f = f \circ T_B$ almost everywhere, and ergodicity makes f constant. If that constant is positive, the whole measure is absolutely continuous. But an absolutely continuous invariant probability has Fourier coefficients $\hat{\lambda}(n) = \hat{\lambda}(Bn)$, which tend to zero for $n \neq 0$ by the Riemann–Lebesgue lemma. It is therefore Haar. Since the present law is not Haar, it is singular. The inverse image in $\mathbb{R}$ of a null set of full λ -measure is a countable union of null translates, proving singularity of the real law in (30).

Finally, there are infinitely many levels with $r_h > 0$. Select one raw digit at each such level and put zeros at all other positions. The resulting base- p random expansion is atomless: every real number has at most two base- p expansions with digits $0, \dots, p-1$, and every fixed infinite choice at the selected positions has probability zero. The remaining digits are independent of this selected subseries. Convolution with an atomless measure remains atomless, since the mass of a singleton is the integral of singleton masses. This proves the assertion for W and hence for S . $\square$

Lemma 9.2 (A compact probability has no vanishing positive-order derivative). *If ρ is a compactly supported probability measure and $m \geq 1$, its m th distributional derivative is not the zero distribution.*

Proof. If it vanished, its Fourier transform would give $(-it)^m \hat{\rho}(t) = 0$ for all real t in our sign convention. Thus $\hat{\rho}(t) = 0$ for $t \neq 0$, contradicting continuity and $\hat{\rho}(0) = 1$. $\square$

Theorem 9.3 (Exact critical regularity). *Let an admissible pure-power source have slack g_h periodic for $h \geq H$. Set*

$$m = \min_{h \geq H} g_h, \quad r_h = g_h - m \quad (h \geq H), \quad a = p^{1-H}.$$

Let P be the finite atomic law of the prefix digit sum at levels $h < H$. Then the residual has the factorization

$$\rho = P * \mathcal{U}_a^{*m} * \sigma, \quad (31)$$

where σ is the tail law made from the r_h digits. Empty convolution powers mean δ_0 . Exactly the following alternatives hold.

- (i) If $r_h = 0$ throughout the periodic tail and $m = 0$, then $\rho = P$ is finite atomic.
- (ii) If $r_h = 0$ throughout the tail and $m \geq 1$, then ρ is a finite mixture of compact splines. For $m = 1$ it has a piecewise constant density but no continuous density. For $m \geq 2$ its density belongs to $C^{m-2}(\mathbb{R})$ and not to $C^{m-1}(\mathbb{R})$.
- (iii) If the periodic tail r_h is not identically zero and $m = 0$, then ρ is singular continuous.
- (iv) If the periodic tail r_h is not identically zero and $m \geq 1$, then ρ has a density in $C^{m-1}(\mathbb{R})$ and not in $C^m(\mathbb{R})$.

These regularity statements concern existence of a globally continuous representative of the density and its derivatives, not arbitrary changes of a density on null sets.

Proof. Separate the finite prefix in theorem 7.2, and remove m copies of a complete infinite digit tail. By theorem 7.1, each complete tail starting at H is uniform on $[-p^{1-H}, p^{1-H}]$. This proves (31). If r is nonzero, its periodic minimum is zero, and rescaling and reindexing the tail puts it under theorem 9.1. Thus σ is singular continuous.

The distributional derivative of the uniform measure is

$$D\mathcal{U}_a = \frac{\delta_{-a} - \delta_a}{2a} =: B_a. \quad (32)$$

If $\sigma = \delta_0$, the residual is a finite sum of translates of an m -fold uniform convolution. For $m \geq 2$, each distributional derivative through order $m - 2$ is a finite sum of translates of a convolution retaining at least two uniforms, and hence has a continuous density. This gives a C^{m-2} density. Its derivative of order $m - 1$ is a compactly supported piecewise constant function, because $D^{m-1}\rho = P * B_a^{*(m-1)} * \mathcal{U}_a$. If the original density were C^{m-1} , this piecewise constant derivative would have a continuous representative and would therefore be identically zero. That contradicts theorem 9.2. The same compact piecewise-constant argument applied to the density itself handles $m = 1$. The case $m = 0$ is immediate.

Now suppose σ is singular continuous. Convolution with the finite atomic P preserves both singularity and absence of atoms, proving (iii). For $m \geq 1$, the law $\mathcal{U}_a * \sigma$ has the continuous density

$$x \mapsto \frac{F_\sigma(x+a) - F_\sigma(x-a)}{2a},$$

where F_σ is continuous because σ is atomless. Equation (32) shows that the distributional derivative of ρ through order $m - 1$ has a continuous density, so ρ has a C^{m-1} density. Its next derivative is the finite signed measure

$$D^m \rho = P * B_a^{*m} * \sigma.$$

This measure is singular, being a finite signed sum of translates of a singular measure, and is nonzero by theorem 9.2. A C^m density would make the same derivative absolutely continuous, which is impossible. This proves (iv). $\square$

Why the classification is effective. For finitely many pure-power streams, the certificate already supplies a period L and a valid start $H = A + 1$. Computing the L tail values of g_h determines their minimum m and whether all values equal that minimum. The four-way classification thus needs only integer arithmetic after the existence test. A shorter actual period may be used but is not required. In contrast, no exact critical-regularity classification for arbitrary prime-to- p parts is claimed here.

10 Explicit factorizations and diagnostic examples

10.1 Two identical source streams and a Cantor residual

Theorem 10.1 (A singular residual from two identical geometric factors). *For every prime p ,*

$$X_p \stackrel{d}{=} \frac{1}{p}X_{p^2}^{(1)} + \frac{1}{p}X_{p^2}^{(2)} + Z_p, \quad Z_p = \sum_{j=0}^{\infty} p^{-(2j+1)} D_j, \quad (33)$$

with independent terms and independent centered p -digit variables D_j . The residual Z_p is singular continuous and has Hausdorff dimension $1/2$. Its extreme support points and variance are

$$\pm \frac{p}{p+1}, \quad \text{Var}(Z_p) = \frac{p^2}{3(p^2+1)}. \quad (34)$$

Proof. Both source streams have start 1 and step 2. Their slack is $g_h = 1$ at odd levels and 0 at even levels, which satisfies the Hall inequalities. The canonical digit formula is exactly (33); singular continuity follows from theorem 9.1. The extreme radius is

$$(p-1) \sum_{j \geq 0} p^{-(2j+1)} = \frac{p}{p+1}.$$

Since $\text{Var}(D_j) = (p^2 - 1)/3$, summing the independent variances gives (34).

The residual obeys the iterated-function identity $Z_p \stackrel{d}{=} D_0/p + Z'_p/p^2$. Its p first-level support intervals are disjoint with positive gaps, and every cylinder of depth N has diameter $[2p/(p+1)]p^{-2N}$ and mass p^{-N} . Covering by these cylinders shows that the support has Hausdorff dimension at most $1/2$. Conversely, for any small radius r , choose N with p^{-2N} comparable to r . An interval of length r meets at most a fixed number of depth- N cylinders, by their uniformly separated geometry. Its mass is at most a constant times $p^{-N} \leq Cr^{1/2}$. Summing this bound over arbitrary interval covers shows the dimension is at least $1/2$. The same bound and the cylinder masses give dimension $1/2$ for the probability measure itself. $\square$

For $p = 2$, D_j is a fair sign, and the formula becomes

$$X_2 \stackrel{d}{=} \frac{1}{2}X_4^{(1)} + \frac{1}{2}X_4^{(2)} + \sum_{j \geq 0} \frac{\varepsilon_j}{2 \cdot 4^j}.$$

The residual is a centered, rescaled middle-half Cantor measure with support endpoints $\pm 2/3$ and variance $4/15$. Each source component has a smooth density, but after both are removed the remaining law is singular. Thus the regularity of an admissible residual is not inferred from the regularity of its source components.

10.2 Every finite differentiability order from two base-four copies

Corollary 10.2 (A tunable exact regularity family). *For every integer $a \geq 1$, two independent copies of $2^{-a}X_4$ jointly divide X_2 . If $a = 1$, their residual is the Cantor law above. If $a \geq 2$, the residual has a density in $C^{a-2}(\mathbb{R})$ and not in $C^{a-1}(\mathbb{R})$.*

Proof. The two deadline streams both have start a and step 2. Their slack begins $g_h = h$ for $1 \leq h \leq a$, then alternates between $a - 1$ and a . It is nonnegative, so the factorization exists. The periodic tail has minimum $m = a - 1$ and is not constant. Apply theorem 9.3. $\square$

In particular, two quarter-scaled base-four laws leave a continuous but not continuously differentiable density. Here the exact decomposition is

$$\rho = U_1 * \mathcal{L} \left(\sum_{j \geq 1} 4^{-j} \varepsilon_j \right).$$

Two eighth-scaled copies leave a C^1 but not C^2 density. These exact endpoint regularity obstructions arise from nonzero singular distributional derivatives, not merely from an inconclusive upper bound on a Fourier transform.

10.3 Checks across all four critical regimes

Source removed from X_2	First digit slacks	Residual conclusion
$X_4 + X'_4/2$	$0, 0, 0, 0, \dots$	Point mass at zero.
$X'_2/2 + U/2$	$1, 0, 0, 0, \dots$	Fair atoms at $\pm 1/2$.
$X'_2/2$	$1, 1, 1, 1, \dots$	Uniform on $[-1, 1]$; no continuous density.
$X'_2/4$	$1, 2, 2, 2, \dots$	A continuous piecewise linear density, not C^1 .
$(X_4^{(1)} + X_4^{(2)})/2$	$1, 0, 1, 0, \dots$	Singular continuous Cantor law.
$(X_4^{(1)} + X_4^{(2)})/4$	$1, 2, 1, 2, \dots$	C^0 density, not C^1 .
$(X_4^{(1)} + X_4^{(2)})/8$	$1, 2, 3, 2, 3, \dots$	C^1 density, not C^2 .

The finite-spline rows recover familiar finite-head decompositions, and are included as normalization and boundary-case checks rather than as novelty claims. The singular and mixed smooth–singular rows demonstrate the additional behavior introduced by simultaneous streams.

10.4 A subcritical check and a rejected critical case

Removing one unscaled X_4 from X_2 leaves $\frac{1}{2}X'_4$. Its stream has $\eta = 1/2$, so $\delta = 1/2$; the sharp coefficient in (27) is $-1/(4 \log 2)$, agreeing with the direct base-four product. On the other hand, two unscaled copies of X_4 fail at $K = 0$ despite $\eta = 1$. Their characteristic-function quotient has a pole at π : the source order there is 2 and the target order is 1. This is the smallest possible exact rejection certificate.

10.5 Why periodicity cannot be dropped from the critical theorem

Example 10.3 (Zero surplus density with a smooth residual). *Remove from X_p exactly the target uniforms whose supply exponents are not perfect squares. The residual is then*

$$Z = \sum_{j=0}^{\infty} p^{-j^2} U_j.$$

The removed source has asymptotic deadline density 1, because the number of unremoved exponents through K is only $\lfloor \sqrt{K} \rfloor + 1$. Nevertheless, the residual is C^∞ . For any fixed N , its first N uniform factors give a Fourier bound $O(|t|^{-N})$, while all remaining characteristic

factors have modulus at most 1. Taking N arbitrarily large and applying Fourier inversion proves smoothness. Its slack is unbounded and nonperiodic. Thus the exact classification in theorem 9.3 depends on periodicity, not on the bare statement that the asymptotic surplus density is zero.

11 Exact computation and a formalization boundary

11.1 The accompanying executable certificates

The file `verify.py` uses Python 3.10 or later and only the standard library. Running

```
python verify.py -output verification.json
```

reproduces the exact checks and the example certificates. Its public routines have the following roles.

Routine	Contract
<code>finite_certificate</code>	For target base p^{power} , returns a sorted slot matching or a least-common-multiple zero-order obstruction.
<code>stream_certificate</code>	Returns a complete finite-window certificate, or a violating deadline including its demand and capacity. Inputs are valuation starts, steps, and optional finite deadlines.
<code>demand</code>	Computes the exact integer expression in (10).
<code>run_tests</code>	Compares certificates with independent finite matching and zero-order checks, verifies moment identities, and writes diagnostic examples.

The distinction between a source denominator and its valuation is explicit: `finite_certificate` takes denominators, whereas `stream_certificate` takes deadlines. Invalid primes, nonpositive denominators, invalid steps, and exceeded resource limits raise exceptions. In particular, a large least common multiple is not silently treated as a mathematical rejection.

The execution accompanying this article produced the following results.

Exact check	Number completed
Finite uniform families	9,100
Finite families accepted	546
Uniform interval partitions	96
Rational moment identities through degree ten	1,056
Finite-plus-geometric families	8,855
Geometric families accepted	1,275
Integer period-drift identities	153,390
Base-six Laurent-polynomial identity	1
Deliberately invalid or resource-limited cases	4

The finite-family tests cover all multisets of length at most four with denominators in $\{1, \dots, 12\}$ for target bases 2, 3, 5, 4, 9. The stream tests cover all multisets of at most three streams with starts $0, \dots, 4$ and steps $1, \dots, 4$, each combined with five finite-deadline patterns. The finite matching oracle uses backtracking rather than the sorted deadline routine. The zero-order oracle checks least common multiples of subfamilies. The stream oracle checks a longer direct window and verifies every returned rejection witness, while separate exact tests check the period recurrence. All uniform moment calculations use rational arithmetic.

These computations are finite implementation tests, not a replacement for the infinite arguments in the article. They do not numerically establish positivity, regularity, or a limit theorem. Those conclusions come from the proofs.

11.2 Existing Lean infrastructure and what remains to be formalized

The inspected file `GeneralizedZeroDivisor.lean` already supplies a precise dyadic zero-divisor interface for weighted products. In particular, the source contains the declarations

```
Fabius.generalizedRvachevProduct_eq_zero_iff_int
Fabius.analyticOrderAt_generalizedRvachevProduct_int
```

with the summability and nonzero-integer hypotheses stated in that file [6]. The first identifies the arithmetic zero set; the second identifies its multiplicity. The report-local README also explicitly distinguishes these adjacent zero-divisor theorems from a formalization of the divisor classification itself [5].

This manuscript did not compile the repository or invoke Lean. It therefore makes no claim that its new theorems are already kernel checked. A realistic formalization sequence would be:

1. Establish the finite nested matching equivalence between prefix counts and sorted deadlines, including a proof-producing rejection witness.
2. Extend that matching to countably many demands by the finite cardinality of every bounded-deadline set; record the bound on the sum of matched widths.
3. Formalize the one-uniform digit split and its independent countable product construction, so that positivity is obtained from probability semantics.
4. Connect analytic zero order to the least-common-multiple obstruction, using the existing dyadic interface first and then general prime powers.
5. Prove the geometric deadline recurrence and verify the finite certificate checker against its universal quantified specification.
6. Formalize the vector zero-hyperplane obstruction and the rowwise reduction. The regularity and ergodic arguments should be a separate final development, not hidden as assumptions inside the factorization theorem.

The main existence theorem can be formalized without the critical singularity or smoothness theory. Keeping that dependency boundary explicit makes an initial verified implementation smaller and its trust story clearer.

12 Further research questions and proposed projects

The following are open directions *within the development of this article*. They are not assertions that no corresponding theorem exists elsewhere in the literature. Each question specifies what the present results do and do not settle.

- Q1. Positive factorization for non-prime-power targets.** For integer B with at least two distinct prime divisors, characterize all uniform-convolution factors of μ_B . The base-six example proves that zero multiplicity is insufficient. A promising exact formulation should combine prime-vector divisibility with positivity of finite cyclotomic quotient masks, rather than simply replacing v_p by the largest exponent of B dividing an integer. The first useful milestone is a complete decision procedure for a fixed small number of source uniforms.

- Q2. The full convolution-factor monoid.** The main theorem classifies factors that are themselves sums of independent uniforms. It does not classify every compact probability factor of μ_p . The digit representation exhibits many purely discrete and singular factors outside that input class. Is there a canonical factor description in terms of integer-valued digit multiplicities and finite positive masks, and what indecomposable factors must be added? An answer would distinguish a genuine factorization theorem for all laws from the uniform-source theorem proved here.
- Q3. Critical streams with prime-to- p parts.** Existence depends only on valuations, but the residual law remembers the full integers. Determine exact smoothness, singularity, and Fourier dimension when $\delta = 0$ and some source denominators or bases have nontrivial prime-to- p parts. The pure-power residual construction remains a convolution factor, but that alone does not determine the effect of the additional digit measures. The repository's one-stream reciprocal-integer family is an important test case, not a substitute for the simultaneous classification.
- Q4. Nonperiodic capacity profiles.** Find necessary and sufficient regularity conditions directly in terms of the admissible sequence g_h . Example 10.3 shows that zero asymptotic surplus permits a smooth law when the leftover scales are sparse and nonperiodic. Can a bounded but nonperiodic profile produce a smooth residual? Which growth conditions give prescribed sub-log-squared Fourier scales, and what replaces the explicit maximizing sequence in theorem 8.1?
- Q5. Shorter arithmetic certificates.** The finite-window algorithm is exact but can have a large period $\text{lcm}(s_1, \dots, s_r)$. Seek a certificate whose size is polynomial, or at least subexponential, in the binary input length for natural restricted stream classes. Useful possibilities include residue-class compression, rational generating functions for demand, and short proofs that a floor-sum never exceeds its affine capacity. Any speedup should still return a witness when the condition fails.
- Q6. Stability under approximate factorization.** Exact reciprocal integrality and coordinate alignment are brittle conclusions. Quantify the distance from a proposed source to the admissible set when only an approximate convolution identity is known in a specified metric. A useful result must state its dependence on source support, observation bandwidth, and the smallest relevant Fourier zero. In several dimensions, a particularly concrete goal is a lower bound on approximation error caused by a small but nonzero oblique component.
- Q7. Other target hyperplane arrangements.** Replace the independent coordinate target by an infinite box spline whose uniform directions form a nontrivial finite arrangement. The proof of theorem 5.1 suggests that every source direction must coincide with a target zero-hyperplane direction, but the multiplicity accounting can couple several arithmetic chains. Determine when a direction-by-direction matching is sufficient and when additional positivity obstructions, analogous to base six, appear.
- Q8. Dimensions and fine regularity of critical digit laws.** For periodic excess profiles taking only the values 0 and 1, separated digit constructions give accessible dimensions. Repeated digit layers introduce carries and overlaps. Compute exact dimensions and local dimensions for these profiles, and determine fractional Hölder, Sobolev, or Besov regularity after the uniform smoothing layers are added. The present theorem settles exact integer C^k thresholds but not these finer scales.
- Q9. Certified construction and formal verification.** Implement a proof-producing checker for finite and geometric demands, then connect its output to a Lean probability-

law construction. The natural initial endpoint is the prime-base existence equivalence, not every analytic corollary. A later certified sampler could use the uniform residual-tail bound (8) to return a sample coupled to the exact residual with a rigorous error radius. This would connect arithmetic certificates, probability semantics, and practical computation in a single development.

13 Conclusion

For prime-base Fabius laws, a simultaneous factorization question in probability reduces exactly to a nested arithmetic allocation problem. Every source uniform must have reciprocal-integer width, and the number of sources whose eligible slots end by level K cannot exceed $K + 1$. The least-common-multiple zero obstruction proves necessity, while a uniform digit split proves sufficiency and supplies the residual constructively.

The consequences go beyond an existence test. Geometric source families admit finite certificates. Multidimensional sources cannot use oblique directions, and exact linear copy-factors form a countable matrix semigroup with no nonconstant continuous paths. A prime-power extension and a base-six signed quotient identify the boundary between an analytic divisor and a positive convolution factor. Finally, unused digit capacity determines the residual's regularity: positive linear growth gives sharp log-squared Fourier decay, while periodic critical capacity gives an exact atomic-spline-singular-smooth classification.

The mathematical core is self-contained and does not assume the correctness of unverified repository research claims. Its most direct next step is a verified matching-and-splitting implementation, followed by the harder classification of positive factors beyond prime-power target bases.

A Source provenance and the scope of the review

The repository’s main tree was recorded at commit

23dd71d2d3d5e85d2b97a8cc45f55e735d69a4ae.

The root and Fabius READMEs, the representation source audit, and the specific Lean zero-divisor source were inspected at that pinned snapshot. GitHub’s indexed code-search results supplied the existing reciprocal-integer divisor report and its README at a second explicit snapshot,

e9c09b12549ea4e3bea7cebc52b450b761f6f581.

These are two identified source versions, not a claim that all inspected material came from one synchronized checkout. The mathematical baseline for the single-copy result is the report cited in [4].

The review deliberately checked for a collision with the proposed scalar factorization idea. The existing report was found, and the scalar theorem was therefore retained only as background. The article’s focus shifted to the full simultaneous uniform-family criterion, multiple geometric streams, the matrix classification, and the residual capacity profile. The source review does not establish an exhaustive absence result for these extensions across every current or historical ProveIt document. Likewise, the external references establish background and terminology, not a global priority certificate.

The external sources used substantively were Arias de Reyna’s account of the compactly supported smooth function, Hall’s original representative-selection paper, and Jurek’s discussion of Urbanik and Mehler semigroups. The proofs here supply the particular matching, positivity, geometric, and regularity statements rather than assigning them to those references.

References

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- [4] ProveIt research corpus, *Reciprocal-Integer Convolution Divisors of the Rvachev Law: Arithmetic semigroups, singular-to-smooth parity, Thue–Morse quotients, zero zeta functions, and inverse-Fabius approximants*, report dated 30 August 2026. Pinned repository source (e9c09b12549e).
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